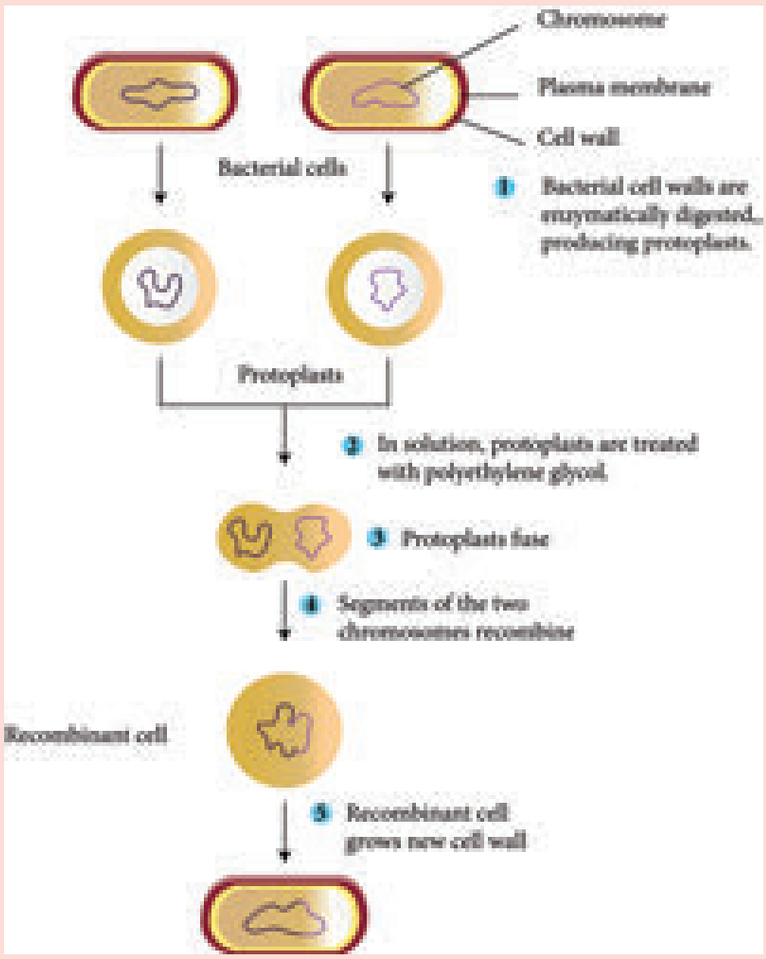
Attributes of Improved strains

1. Assimilate inexpensive and complex raw materials efficiently.
2. Alter product ratios and eliminate impurities or by products in downstream processing.
3. Reduce demand on utilities during fermentation (air, cooling water, or power).
4. Provide cellular morphology in a form suitable for product separation.
5. Create tolerance to high product concentration.
6. Shorten fermentation times.
7. Overproduce natural products or bioactive molecules not synthesized naturally for example insulin.
8. Excrete the product to facilitate product recovery.

Generally wild strains of microorganisms produce low quantities of commercially important metabolites. So, genetic improvements have to be made and new strains need to be developed for any substantial increase in the product formation in a cost effective manner.



Protoplast fusion is defined as fusion of two different protoplasts. A cell wall less nature of plant, bacteria, fungal is called protoplast. It is re moved by either mechanical or enzymatic means. Protoplast has nucleus other protoplasmic contents which are surrounded by cytoplasmic membrane.



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